

Laporan Progress Project Material Incoming

Tanggal: 2026-09-05

Repository: material_incoming

Branch: main

Ringkasan

Project ERP Material Incoming telah melalui tahap refactoring besar dari arsitektur legacy ke NewSchema domain-driven design. Semua modul utama sudah berjalan di atas struktur baru, dengan fokus pada stabilitas traceability, produksi, dan integrasi subcon.

Capaian Utama (Agustus–September 2026)

Modul	Status	Keterangan
Incoming (Arrival/Receive)	Stabil	Create arrival, receive, QR label, print invoice, import dokumen, dropdown size diperbaiki.
Inventory	Stabil	Stock card, location stock, inventory movements, transfer, adjustment, stock opname.
Production	Stabil	WO, routing, hourly reports, backflush, material issue, handover.
Planning (Forecast & MRP)	Stabil	Forecast list, MRP dengan family filter, pivot table bulanan.
Outgoing	Stabil	DO, DN, picking, trucking, standard packing.
Subcon	Stabil	PO subcon, receive subcon, traceability.
Administrasi	Stabil	User & role management, permission.
QR Label / Traceability	Diperbaiki	Batch identity sekarang konsisten (tag == batch_no).
Part Master	Stabil	CRUD GCI Part, vendor part, BOM, substitute, subcount mapping.

STATISTIK

- **Routes:** 99+ (incoming, inventory, master_data, outgoing, planning, production, purchasing, subcon, warehouse)
- **Controllers:** ±80 file
- **Models:** 70+
- **Migrations:** 289
- **Commands:** 8
- **Services:** 7

Alur End-to-End (Incoming Material → FG Dispatch)

1. INCOMING MATERIAL (RECEIVING)

1. **Create Arrival** – User membuat kedatangan (arrival) dengan data vendor, kontrak, item (part, qty, uom).
2. **Receive** – Setelah barang fisik tiba, user melakukan receive per item. Sistem menghasilkan QR label (tag RCV-xxx) sebagai identitas batch.
3. **Putaway** – Stock masuk ke lokasi gudang (`inventory_location_stock`) dengan `batch_no = tag` , dan mencatat riwayat gerak di `inventory_stock_movements` .
4. **Quality Inspection** (Opsional) – Inspeksi kualitas dicatat di `arrival_inspections` dan `container_inspections` .

2. INVENTORY MANAGEMENT

- Stock tersedia di `inventory_location_stock` dengan batch dan lokasi spesifik.
- Monitoring via **Stock Card** (per part, per lokasi, per batch).
- Transfer antar lokasi dan adjustment tersedia.
- Stock Opname periodik.

3. PLANNING (FORECAST & MRP)

1. **Forecast** – User input forecast demand (per part, per bulan) melalui `ForecastDocument` dan `ForecastDocumentRow` .
2. **MRP** – Sistem menjalankan MRP berdasarkan forecast, menghasilkan:
 - **Purchase Plan** (BUY) untuk RM.
 - **Production Plan** (MAKE) untuk FG/WIP.
3. **Family Filter** – MRP dan Forecast bisa difilter berdasarkan product family (Comp Base, Back Plate, Reinforce, Tray Drip, Small Part) dari `part_name` .

4. PRODUCTION (WORK ORDER)

1. **Material Request** – Berdasarkan MRP, sistem membuat material request ke gudang.
2. **Warehouse Supply** – Gudang menyuplai material ke produksi.
3. **Start Production** – WO dimulai dengan operator, mesin, dan proses.

4. **Hourly Reports** – Operator mencatat output aktual (`qty_actual` , `qty_ng`) per jam.
5. **Backflush** – Konsumsi material otomatis saat output FG dicatat, mengurangi stock lokasi.
6. **Handover** – Transfer WIP antar proses.
7. **Finish WO** – WO selesai, FG masuk stock.

5. OUTGOING (DELIVERY & PICKING)

1. **Delivery Order (DO)** – Daftar permintaan pengiriman dari customer.
2. **Picking** – Operator scanning lokasi dan part, mengurangi stock FG.
3. **Delivery Note (DN)** – Surat jalan dan packing list dicetak.
4. **Trucking** – Pengiriman dijadwalkan.
5. **Shipment** – Barang keluar, stock berkurang.

6. SUBCON (OUTSOURCING)

- Subcon PO dibuat untuk parts yang diproses di vendor luar.
 - Material dikirim ke subcon.
 - Hasil jadi diterima kembali, dicatat batch dan stock.
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Fitur Unggulan

INCOMING & INVENTORY

- Import dokumen (Excel) untuk receive.
- Cetak label QR untuk setiap receive.
- Print invoice, inspection report, detail arrival.
- Stock card dengan pergerakan per batch.
- Multi-lokasi dan multi-batch.
- Stock opname dengan barcode scanning.
- QR label konsisten sebagai `batch_no` .

PRODUCTION

- BOM aktif dengan routing (proses, mesin, material).
- Work Order dengan status (released, in_production, paused, material_hold, dll).
- Operator board per mesin.
- Hourly reporting dengan breakdown `actual`, `ng`, `ng_reason`.
- Material issue & handover antar proses.
- Backflush konsumsi material.
- Support WIP stock antar proses.
- Integrasi dengan ProductionGciApi (Android app).

PLANNING

- Forecast list dengan pivot bulanan.
- MRP dengan family filter dan badge.
- History audit.

OUTGOING

- Delivery Order, Delivery Note, Picking FG.
- Print surat jalan, packing list, invoice.
- Support standard packing.

ADMINISTRASI

- User & role management dengan permission granular.
 - Dashboard dan logs.
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Perbaikan Terbaru (September 2026)

Commit	Deskripsi
019e029	fix(arrivals): include gci_parts.size in API and prioritize it in size dropdown
8efb56c	fix(arrivals): size dropdown uses register_no before part_no fallback
ee9c80e	fix(production): drop nonexistent batch_no predicate in backflush traceability
e04627a	fix(inventory): backfill receive tag onto existing location-stock batch
ca236a4	fix(production): backflush movement lookup uses from_location_code
17c6afa	fix(production): write production order number as FG stock batch
688452e	fix(subcon): write subcon order_no as batch_no on receive
4d16664	fix(receive): write receive tag as batch_no on putaway/issue
10c11ec	feat(inventory): add identity columns to inventory_stock_movements
07ce7d2	fix(arrivals): show part in size dropdown when register_no is blank
36c2776	fix(receive): qualify topVendors columns with correct table name
87cdefb	fix(inventory): use NewSchema part accessors in receives view
4828053	fix(incoming): resolve soft-delete alias conflict in receives queries
ca16a4e	fix(incoming): restore display_part_no/name accessors on NewSchema item
e0cf053	feat(incoming): add TRIAL INVOICE xlsx import command
e69eb6f	feat(planning): add family filter + badge to MRP module
cdd4e0c	feat(planning): pivot forecast table to one row per part with month columns
480a08d	feat(planning): switch period filter to month picker + clear month labels

Arsitektur Teknis

DOMAIN MODELS (NEWSHEMA)

- **Core** : GciPart, Vendor, VendorPart, Customer, Uom, Machine, dll.
- **Incoming** : IncomingArrival, IncomingArrivalItem, IncomingReceive.
- **Inventory** : InventoryLocationStock, InventoryStockMovement, InventoryTransfer, StockOpname.
- **Production** : ProductionOrder, ProductionGciHourlyReport, ProductionGciDowntime, Bom, BomItem.
- **Outgoing** : OutgoingDeliveryNote, OutgoingDeliveryOrder, OutgoingPo, PickingFg.

POLA UMUM

- **Stock Management**: `InventoryLocationStock::consumeStock()` dengan named arguments.
- **Traceability**: `batch_no` pada stock movement sama dengan QR tag.
- **Backflush**: konsumsi material otomatis saat output FG dicatat.

DEPENDENCIES

- Laravel 12
 - MySQL (legacy) + PostgreSQL (NewSchema target)
 - QR Code generator
 - Excel (import/export)
 - Pusher (broadcast monitoring)
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Status Testing & QA

Area	Status	Catatan
Unit Test	Partial	Belum ada test suite penuh.
Integration Test	Manual	Pengujian end-to-end manual.
QA Regression	Pass	Q-014 (incoming arrival) sudah fix.
UI/UX	Good	Perbaikan dropdown dan aksesibilitas.

Known Issues / Open Items

- **Testing:** Belum ada CI/CD.
- **Dokumentasi:** Endpoint API belum terdokumentasi (OpenAPI/Swagger).
- **Performance:** Query MRP dan forecasting perlu optimasi indexing.
- **Subcon:** Fitur pengembalian material masih dalam tahap perbaikan.
- **User Permissions:** Beberapa modul baru belum memiliki permission lengkap.

Next Steps

1. **Penambahan Test Suite** – Unit test untuk service layer dan integrasi.
2. **Optimasi Query** – Index di `inventory_location_stock`, `production_orders`, `mrp_*`.
3. **Dokumentasi API** – Integrasi dengan Laravel Swagger atau Postman Collection.
4. **Penyempurnaan Subcon** – Selesaikan alur return dan traceability.
5. **Monitoring Real-time** – Perbaiki broadcast untuk operator board.
6. **Training User** – Siapkan user manual dan video tutorial.

Kesimpulan

Project Material Incoming mencapai tahap **stabil** dengan semua modul utama berfungsi di atas NewSchema. Fix terakhir untuk traceability batch dan UI dropdown sudah selesai.

Progress overall: ~85%

Target Go-Live: Akhir September 2026 (tentatif)

Laporan dibuat berdasarkan commit history, route/controller structure, dan dokumentasi internal project.